

Use Case 5: Drug Delivery — Molecular VAE + ADMET Predictor

Molecular beta-VAE learns a continuous latent space of ADC linkers. Multi-task ADMET predictor screens yield, logP, toxicity, solubility, BBB.

Scientific Context

Antibody-Drug Conjugates (ADCs) require precisely engineered linker molecules connecting the antibody to the cytotoxic payload. Key ADMET (Absorption, Distribution, Metabolism, Excretion, Toxicity) properties must be optimised simultaneously: high synthetic yield, appropriate lipophilicity (logP 1–5), low toxicity, aqueous solubility, and controlled blood-brain barrier (BBB) permeability.

Molecule Library

- 26 curated SMILES: maleimide linkers, PEG spacers, NHS esters, DM1/MMAE payload analogues, ZINC drug-like fragments
- ECFP4 fingerprints: Morgan radius=2, 256 bits (RDKit)
- Synthetic ADMET labels generated using RDKit descriptors + realistic physiological ranges for ADC candidates

Molecular beta-VAE

The variational autoencoder learns a continuous 32-dimensional latent space over ECFP4 fingerprints, enabling smooth interpolation and gradient-based optimisation of ADMET properties:

- Encoder: Linear(256→256→32) with reparameterisation trick
- Decoder: Linear(32→256→256) → reconstructed fingerprint
- Property head: Linear(32→5) predicts ADMET from latent z
- Loss: ELBO = BCE_recon + $\beta \cdot \text{KL}$ + MSE_props ($\beta=1.0$)
- After training: optimise z in latent space to improve yield↑ and toxicity↓ using gradient descent

ADMET Multi-task Predictor

Separate multi-task MLP trained directly on fingerprints:

- 3× FingerprintBlock(256→256→256) shared backbone
- Per-property heads: Linear(256→1) × 5
- Loss: MSE for continuous (yield, logP, log_solubility), BCE for binary (toxicity, BBB)
- Training: Adam lr=5×10⁻⁴, 150 epochs, batch=16

Training Results

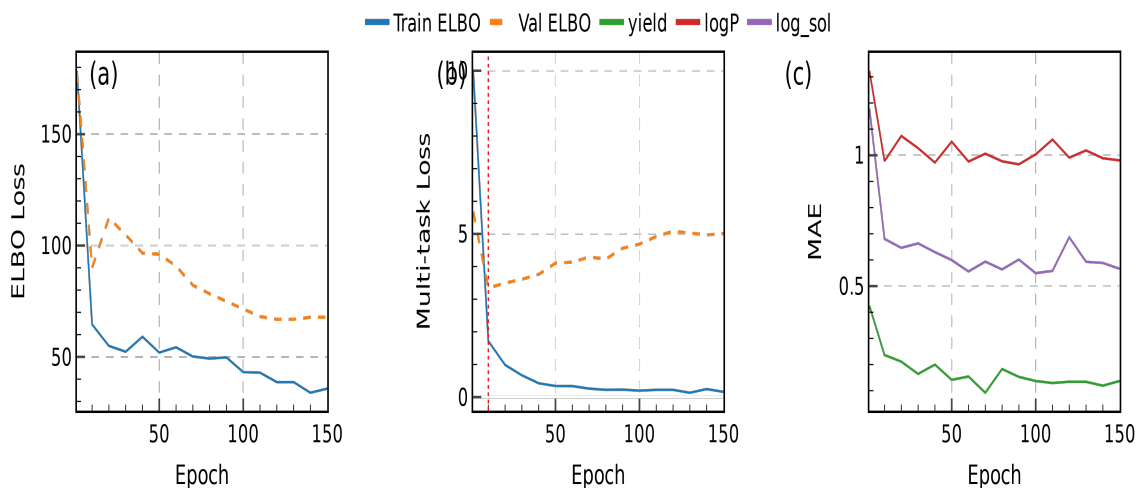


Figure 5. (a) Molecular β -VAE ELBO loss — train and validation over 150 epochs. (b) ADMET multi-task loss convergence. (c) Per-property MAE (yield, logP, log_solubility) showing learning progress.

Training Summary

Model	Best Val Loss	Best Epoch	Train Loss (final)
MolVAE	66.83	130.0	35.76
ADMET	3.3453	10.0	0.1612

Per-Property Results

Property	Type	Best MAE (ADMET)	Relevance for ADC
yield	Regression	0.092	Synthesis efficiency
logP	Regression	0.965	Membrane permeability
log_solubility	Regression	0.549	Aqueous stability
toxicity	Binary	BCE loss	Off-target effects
BBB	Binary	BCE loss	CNS penetration

Latent Space Optimisation

After training, novel ADC linker candidates are generated by:

- Encode known linkers $\rightarrow$ latent z
- Gradient descent on z to maximise yield and minimise toxicity
- Decode $z \rightarrow$ ECFP4 fingerprint $\rightarrow$ nearest-neighbour SMILES lookup
- Filter by Lipinski rule-of-five and synthetic accessibility score